Anthony Hana

anthonyhana04@gmail.com | anthonyhana.com | github.com/anthonyhana04 | linkedin.com/in/anthony-hana

Education

McMaster University **Expected Dec 2027**
Honours Mathematics and Computer Science (B.Sc.) *Hamilton, ON*

- **Related Courses:** Data Structures & Algorithms, Algorithms and Complexity, Databases, Computer Architecture, Natural Language Processing, Linear Algebra, Computer Graphics.

Experience

Data Engineer Intern **Jun 2026 - Aug 2026**
Kroll *Toronto, ON*

- Built a unified client dataset in Databricks from **3** enterprise sources (Databricks, Dynamics, D&B), resolving duplicates and inconsistencies across **2,000+** clients for analysts.
- Implemented a Databricks/Azure dashboard monitoring **300+** running jobs by compute type (serverless vs. classic) and runtime, letting engineers stop long-running jobs and switch compute types to cut costs.

Software Engineer Intern **Sep 2025 - May 2026**
LTV.ai *Toronto, ON*

- Solely built a self-serve analytics feature adopted by all customers, replacing manual SQL reports written by engineers and helping sign and renew **$3M+ ARR** in contracts.
- Cut dashboard load times from **30 minutes to seconds** across **1.1B+** rows by building ClickHouse materialized views with late-event handling and deduplication, helping retain **4** at-risk customers.
- Built a RAG system that drafts marketing emails from customer data, past emails, and brand documents for users to edit and send, saving customer service teams **5+ hours** of copywriting each week.
- Increased email sending throughput **4x** by adding caching and fallbacks for missing customer data to the AI pipeline, and added end-to-end logging for visibility into each stage.

VP Software Architecture **Jun 2025 - Present**
McMaster Start Coding *Hamilton, ON*

- Develop the compiler and its error handling for the Stabl.Rocks IDE, showing **80,000+** students clear errors in their code across workshops and university coursework.
- Cut **26%** of the codebase by removing unused code and deprecated features.
- Develop custom Haskell plugins for McMaster Start Coding's multisite platform and core course workflows.

Projects

LavaLock - DeltaHacks XII | | | Swift, Go, C++, AWS, PostgreSQL

- Won the Swift Student Challenge prize at DeltaHacks XII, out of **144** teams, by building a lava lamp-powered MFA and password vault that draws randomness from physical entropy.
- Built a native SwiftUI password vault and MFA app with Keychain storage and cryptographic password generation seeded by lava-lamp entropy served from the Go/C++ pipeline.
- Created a Go/C++ entropy pipeline on AWS S3/EC2 that collects and serves randomness from lava lamp imagery.

TeX-it - HackThe6ix 2025 | | | Electron, React, TypeScript, Go

- Built a desktop LaTeX IDE with real-time editing and rendering, plus validation that ensures compile-safe output.
- Achieved **96%** accuracy across **10,000** test inputs on compile-ready LaTeX output by developing AI pipelines that convert **3** input types (voice, image, and text).

Technical Skills

Languages: Python, Go, C++, JavaScript, TypeScript, Swift, SQL, Haskell
Frameworks & Tools: Next.js, Node.js, REST APIs, Git, GitHub Actions, Docker, Terraform, AWS
Data: PostgreSQL, Redis, ClickHouse, Kafka, Databricks, Azure, pandas, NumPy
AI & Developer Tooling: Claude Code, Cursor, GitHub Copilot, RAG, LLMOps